Power input and +18V4 SEPIC converter

File: pwr_input.kicad_sch

−18V4 Cuk converter

File: pwr_neg.kicad_sch

+50V boost and +48V phantom supply

File: pwr_phantom.kicad_sch

Linear regulators: +−16V, +5VA, +3V3A

File: pwr_ldo.kicad_sch

Digital supplies: +3V3D, +0V9, +1V8

File: pwr_digital.kicad_sch

USB−C connector and PD sink

File: usb_pd.kicad_sch

XMOS XU316, boot flash, debug

File: xmos.kicad_sch

Audio master clocks, I2S, LEDs

File: clocks.kicad_sch

DAC (PCM1794A)

File: dac.kicad_sch

DAC I/V stage, output filter, volume

File: dac_iv.kicad_sch

Headphone amplifier

File: hpamp.kicad_sch

Output relay, protection, headphone jacks

File: hp_protect.kicad_sch

Microphone input, phantom feed, preamp

File: mic.kicad_sch

ADC driver and ADC (CS5361)

File: adc.kicad_sch

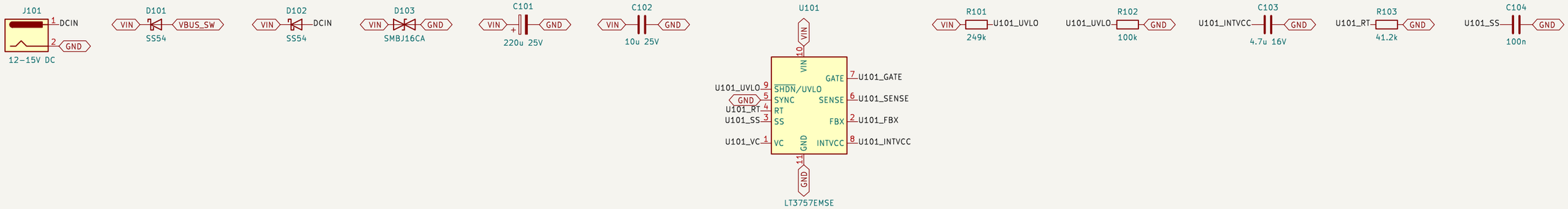
ALL−IN−ONE AUDIO INTERFACE − schematic rev A (generated by hardware/schematic/design.py)
USB−C UAC2 (XMOS XU316) | PCM1794A DAC | headphone amp >= 2 W @ 32R (OPA1612 + 2x LME49600 per ch)
XLR mic input: THAT1512 preamp 0−60 dB, switchable +48 V phantom, CS5361 ADC
Nets are joined by labels: global labels across sheets, local labels within a sheet.
See docs/design.md for the full description and the list of items to verify before layout.

All−in−one audio interface
VincentM9

Sheet: /
File: aio.kicad_sch

Title: All−in−one audio interface

Size: A3	Date: 2026−10−23	**Rev: A**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 1/15



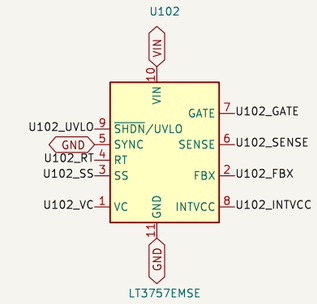
POWER INPUT / +18V4
– VIN = USB-C VBUS (after the STUSB4500 load switch) OR 12–15 V DC jack, Schottky OR, TVS.
– U101 LT3757 SEPIC –> +18V4. FBX = +1.6 V x (1 + 105k/10k). RT 41.2k = 300 kHz.
  Current limit 110 mV / 30 mR = 3.7 A. Feedback is taken before the 1 uH post-filter.
– VERIFY: LT3757 compensation, FET and inductor choices (simulate in LTspice).

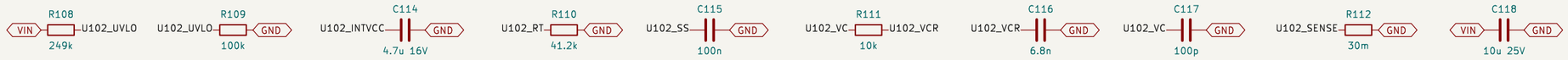
All–in–one audio interface
VincentM9

Sheet: /Power input and +18V4 SEPIC converter/
File: pwr_input.kicad_sch

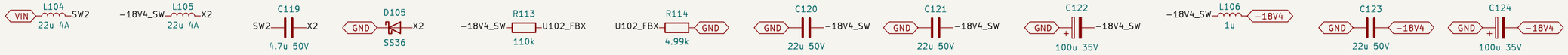
Title: Power input and +18V4 SEPIC converter

Size: A3	Date: 2026–10–23	**Rev: A**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 2/15











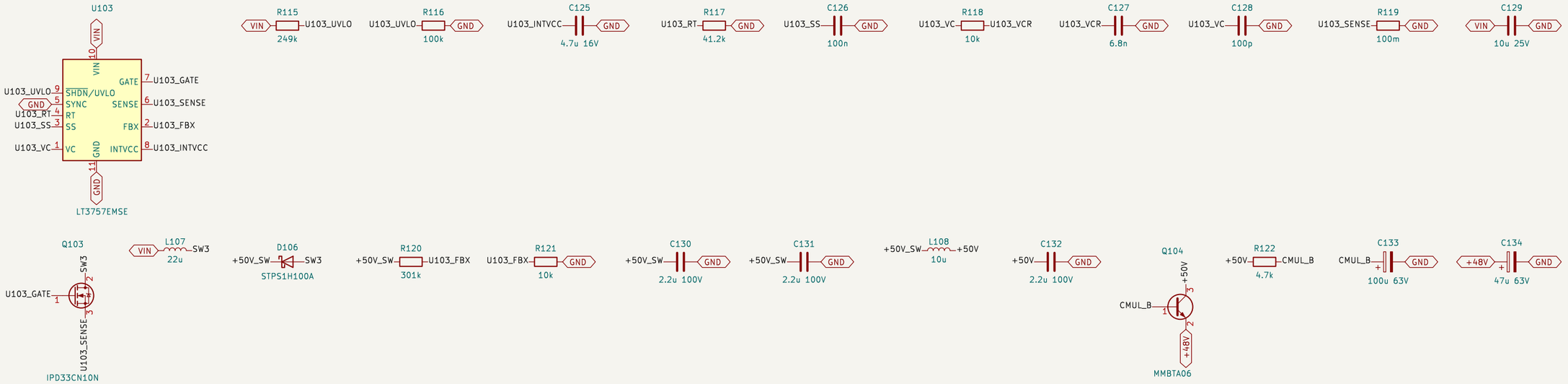
−18V4
- U102 LT3757 Cuk (inverting) -> −18V4. FBX = −0.8 V x (1 + 110k/4.99k).
- Diode anode at the switch-side node X2, cathode to GND; output inductor from X2 to −18V4_SW.

All-in-one audio interface
VincentM9

Sheet: /−18V4 Cuk converter/
File: pwr_neg.kicad_sch

Title: −18V4 Cuk converter

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 3/15



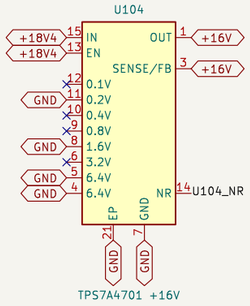
PHANTOM SUPPLY
- U103 LT3757 boost -> +50V (FBX = 1.6 V x (1 + 301k/10k)), 100 V FET and diode.
- Q104 capacitance multiplier -> +48V (about 49.3 V open-circuit, 48.6 V at 15 mA).

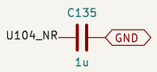
All-in-one audio interface
VincentM9

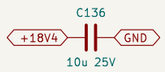
Sheet: /+50V boost and +48V phantom supply/
File: pwr_phantom.kicad_sch

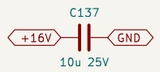
Title: +50V boost and +48V phantom supply

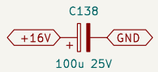
Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/15

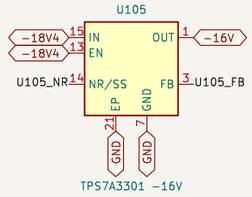


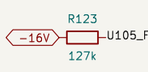


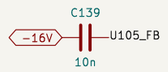


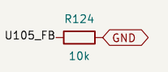


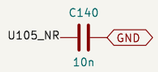


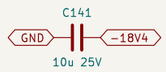


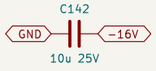


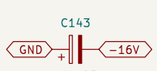


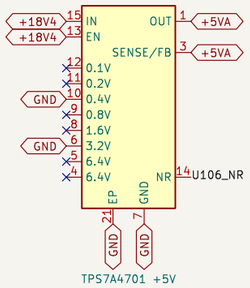


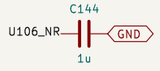


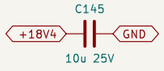


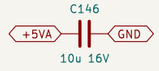


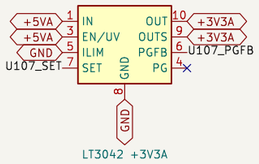




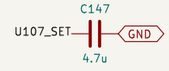


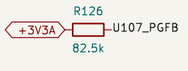




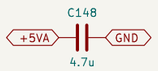


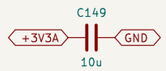












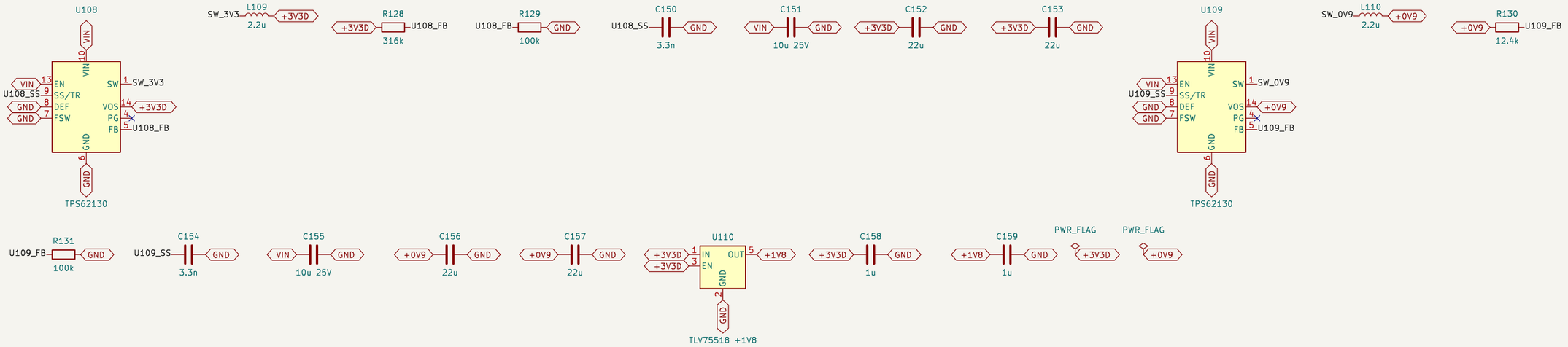
LINEAR REGULATORS
- U104 TPS7A4701 ANY−OUT +16.0 V (6.4+6.4+1.6+0.2 grounded). U106 TPS7A4701 +5.0 V (3.2+0.4).
- U105 TPS7A3301 −16.1 V = −1.176 V x (1 + 127k/10k). Pad = GND, EN = IN (datasheet).
- U107 LT3042 +3V3A (100 uA x 33.2k) for oscillators and converters.
- Put every thermal pad on copper pour; the +−16 V LDOs dissipate ~0.7 W at full power.

All−in−one audio interface
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Sheet: /Linear regulators: +−16V, +5VA, +3V3A/
File: pwr_ldo.kicad_sch

Title: Linear regulators: +−16V, +5VA, +3V3A

Size: A3	Date: 2026−10−23	**Rev: A**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 5/15



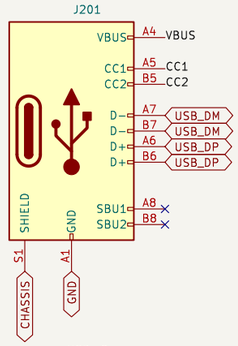
DIGITAL SUPPLIES
- U108 / U109 TPS62130 bucks from VIN -> +3V3D and +0V9 (XMOS core). Vout = 0.8 V x (1 + R1/100k).
- U110 TLV75518 -> +1V8 (XMOS USB PHY).
- VERIFY: TPS62130 FSW pin sense (GND = high frequency?).

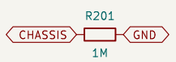
All-in-one audio interface
VincentM9

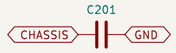
Sheet: /Digital supplies: +3V3D, +0V9, +1V8/
File: pwr_digital.kicad_sch

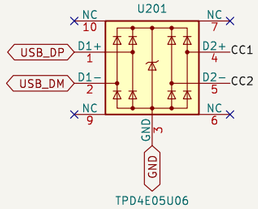
Title: Digital supplies: +3V3D, +0V9, +1V8

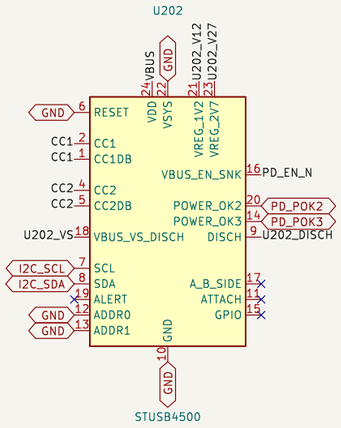
Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/15

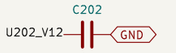


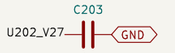


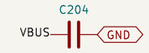


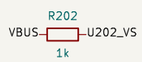


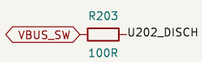












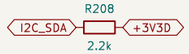














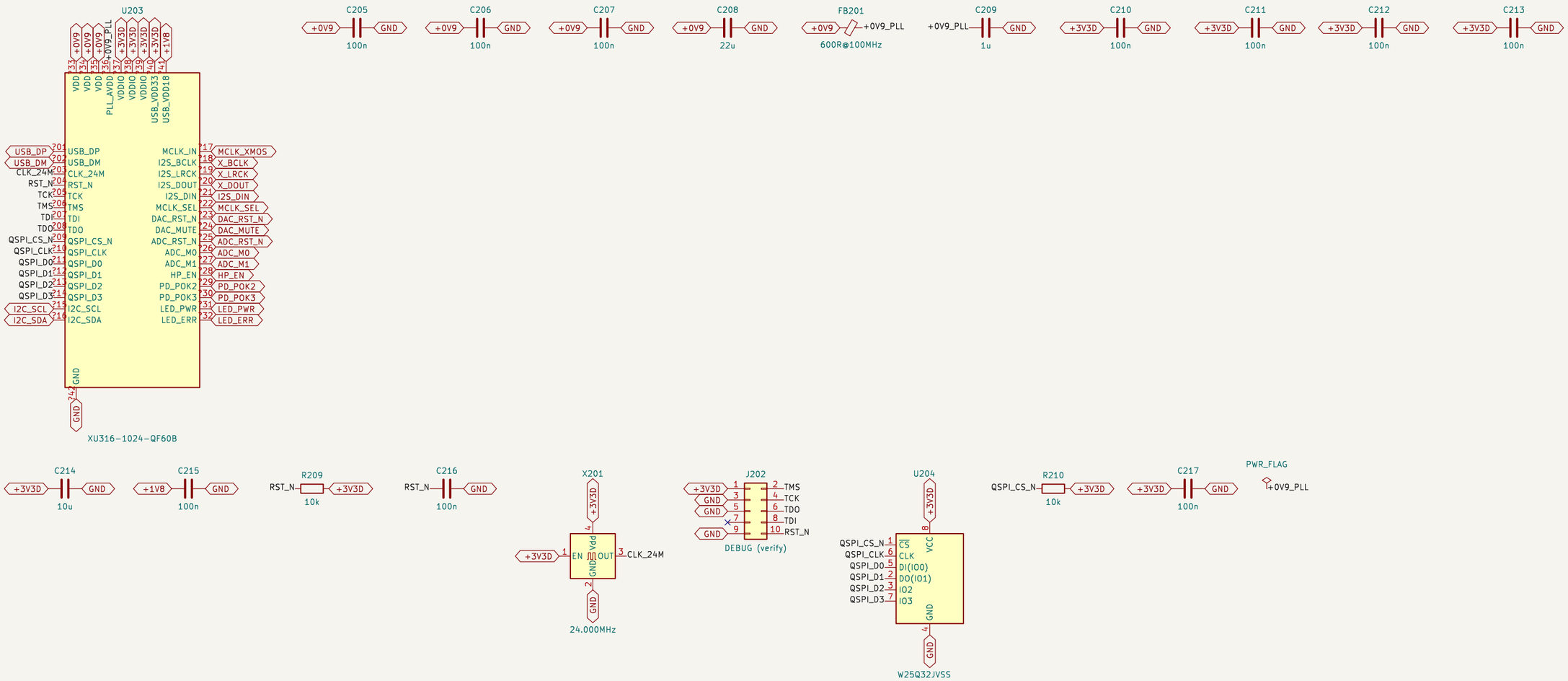
USB-C / PD
- U202 STUSB4500: NVM PDO3 = 15V/1.5A, PDO2 = 12V/1.5A, PDO1 = 5V. VBUS_EN_SNK drives load switch Q201.
- VSYS grounded (unused). DISCH and VBUS_VS_DISCH have current-limiting resistors.
- U201 ESD protection on D+/D-/CC1/CC2. Shield to chassis through 1M || 4.7n.

All-in-one audio interface
VincentM9

Sheet: /USB-C connector and PD sink/
File: usb_pd.kicad_sch

Title: USB-C connector and PD sink

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/15



XMOS
- U203 XU316-1024-QF60B: FUNCTION names only. Pin numbers are placeholders (?nn) and there is no footprint.
  Replace it with the official XMOS library part before layout; assign I2S/GPIO to ports in the .xn file.
  From the datasheet: USB_DM = 28, USB_DP = 29, USB_VDD33 = 30, USB_VDD18 = 31; core 0.855–0.945 V.
- QSPI boot flash on X0D01 / X0D04–07 / X0D10 (check the boot-mode straps in the datasheet).
- J202 debug header pinout is a placeholder: match it to your XTAG adapter.

All-in-one audio interface

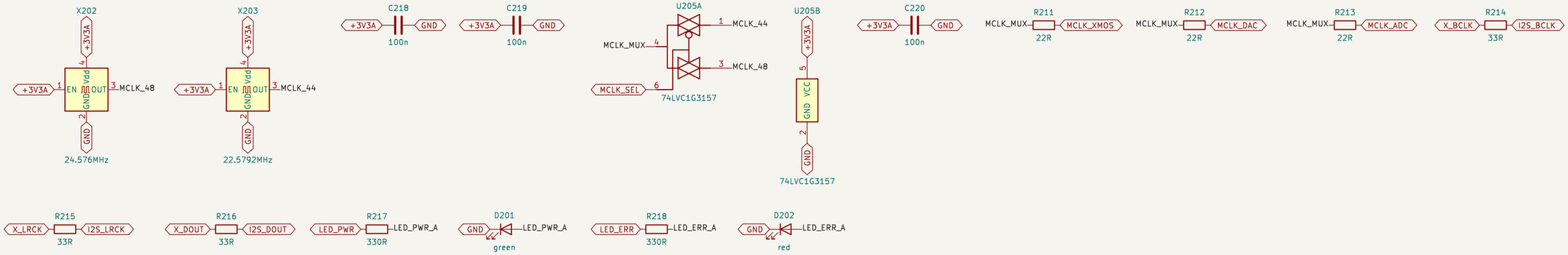
VincentM9

Sheet: /XMOS XU316, boot flash, debug/

File: xmos.kicad_sch

Title: XMOS XU316, boot flash, debug

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/15



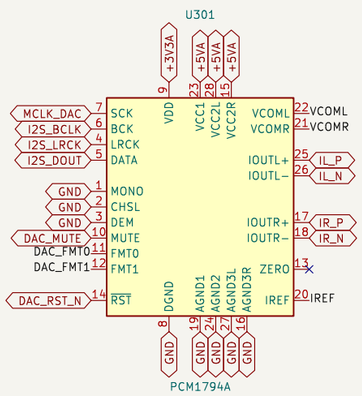
CLOCKS / I2S
- X202 24.576 MHz (48 kHz family) and X203 22.5792 MHz (44.1 kHz family), selected by U205 (MCLK_SEL = 0 -> 48k).
- Mux output fans out through 22R to XMOS, DAC and ADC. XMOS drives BCLK / LRCK / DOUT through 33R.

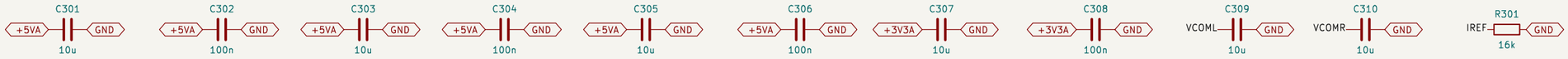
All-in-one audio interface
VincentM9

Sheet: /Audio master clocks, I2S, LEDs/
File: clocks.kicad_sch

Title: Audio master clocks, I2S, LEDs

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/15







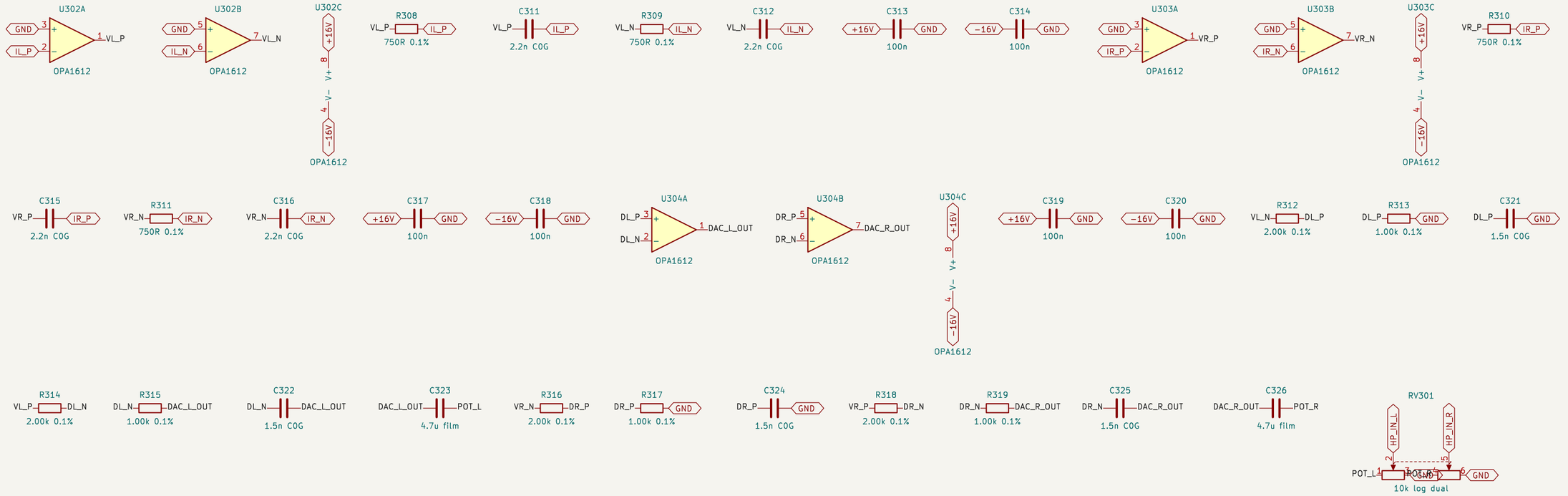
DAC NOTES
- U301 PCM1794A (hardware control). Output 7.8 mA p-p per pin around ~-6.2 mA.
- I/V 750R: 5.85 V p-p per leg around +4.65 V. Diff stage gain 0.5 -> ~2.07 Vrms single-ended.
  Diff stage inputs are swapped (P from IOUT-, N from IOUT+) so the output is non-inverted.
- FMT0/FMT1 straps: fit pull-down OR pull-up per PCM1794A datasheet Table for 24-bit I2S (VERIFY).
- 4.7 uF film caps block DC into the volume pot. Use 0.1 % resistors in the diff stage.

All-in-one audio interface
VincentM9

Sheet: /DAC (PCM1794A)/
File: dac.kicad_sch

Title: DAC (PCM1794A)

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 10/15



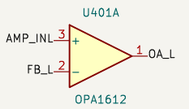
I/V AND OUTPUT
- I/V 750R || 2.2n per leg: 5.85 V p-p around +4.65 V. Diff stage gain 0.5 -> -2.07 Vrms single-ended.
- Diff stage inputs are crossed (P from IOUT-, N from IOUT+) so the output is not inverted.
- 4.7 uF film caps block DC into the volume pot. Use 0.1 % resistors in the diff stage.

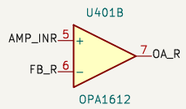
All-in-one audio interface
VincentM9

Sheet: /DAC I/V stage, output filter, volume/
File: dac_iv.kicad_sch

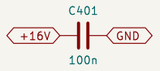
Title: DAC I/V stage, output filter, volume

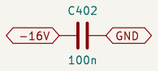
Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/15

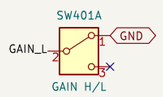


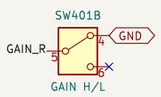




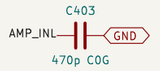






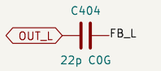


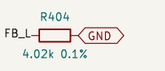


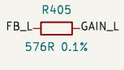


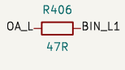


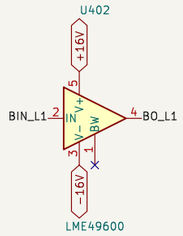


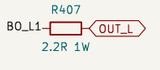


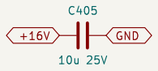


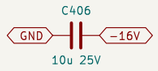




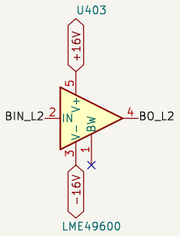


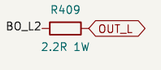


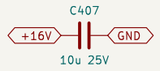


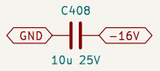


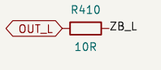


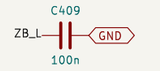


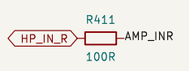


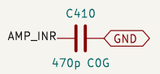


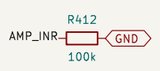


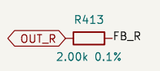


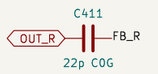


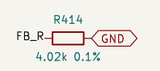


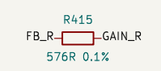




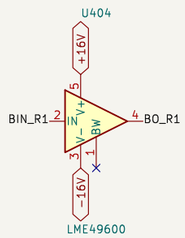


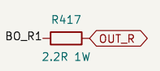


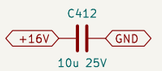


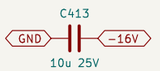


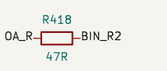


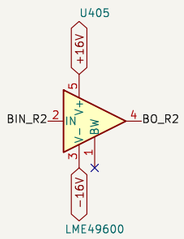


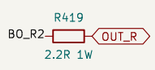


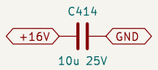


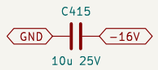


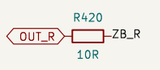


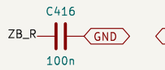


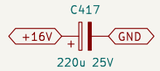


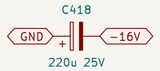








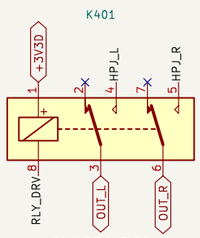




HEADPHONE AMP NOTES
- Per channel: OPA1612 + 2x LME49600 inside the loop, 2.2R ballast each, feedback from OUT_x.
- Gain: 1 + 2.00k / (4.02k || 576R) = 4.97 (SW401 closed, High) or 1.50 (open, Low).
- 2 W into 32R = 8 Vrms, 354 mA peak (177 mA per LME49600, rated 250 mA).
- Relay K401 (NO contacts) connects jacks only when RDRV is high:
  ~2–3 s after power-up (220k / 10u on RDRV), released at once if +18V4 sags (U406A), −16V collapses (U406B),
  either output has DC beyond about +−0.6 V (Q402/Q403), or XMOS pulls HP_EN low (BAT54 D402).
- LME49600 BW pin open = low-power mode (~7.3 mA Iq). Keep each TO-263 tab on >= 4 cm2 copper.

All-in-one audio interface
VincentM9

Sheet: /Headphone amplifier/		
File: hpamp.kicad_sch		
Title: Headphone amplifier		
Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/15





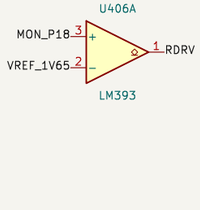


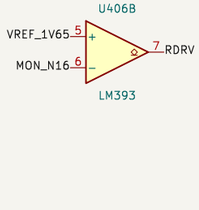






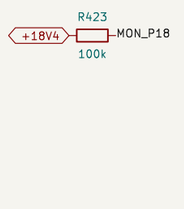


















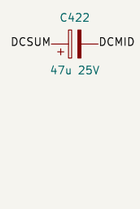


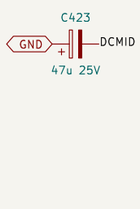


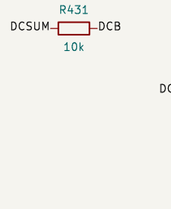




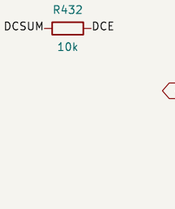






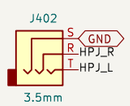












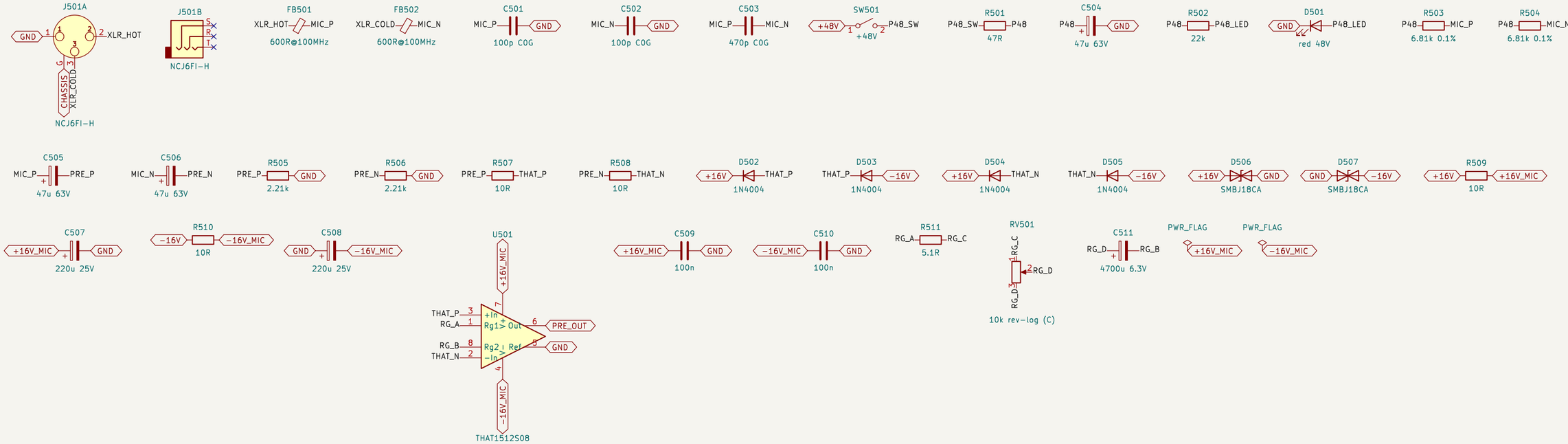
PROTECTION
– Relay K401 (NO contacts) connects the jacks only when RDRV is high: ~2–3 s after power-up (220k / 10u), released at once if +18V4 sags (U406A), −16V collapses (U406B), either output has DC beyond about +−0.6 V (Q402 / Q403), or the XMOS pulls HP_EN low (BAT54 D402).

All-in-one audio interface
VincentM9

Sheet: /Output relay, protection, headphone jacks/
File: hp_protect.kicad_sch

Title: Output relay, protection, headphone jacks

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 13/15



MIC INPUT NOTES
- P48 phantom per IEC 61938: 6.81k 0.1 % matched feed pair, open-circuit ~48.6 V. SW501 switches it.
- Hot-plug protection: 10R series + 1N4004 clamps to the +-16V rails + SMBJ18CA TVS on the rails.
- U501 THAT1512: G = 0.5 + 5k / (5.1R + RV501) -> 0 dB to ~60 dB. 4700u cap blocks DC in the gain leg.
- U502 makes a differential signal: 1 Vrms at PRE_OUT = 2 Vrms differential = 0 dBFS on CS5361
  (full scale = 1.13 x VA = 5.65 V p-p differential at VA = 5 V).
- U503 CS5361 slave mode, I2S, HPF on. M0/M1 speed mode set by XMOS per sample rate.
  Unused right channel tied to VQ. OVFL drives the clip LED.
- VERIFY: CS5361 FILT+/VQ capacitor values and input RC against the datasheet.

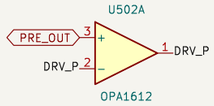
All-in-one audio interface

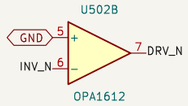
VincentM9

Sheet: /Microphone input, phantom feed, preamp/
File: mic.kicad_sch

Title: Microphone input, phantom feed, preamp

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 14/15

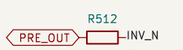


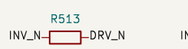








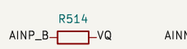










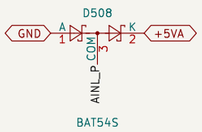


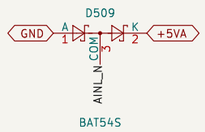


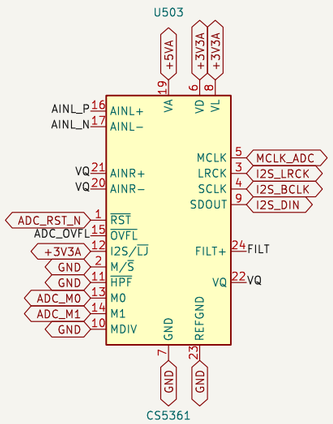










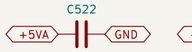


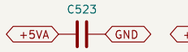


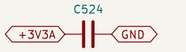


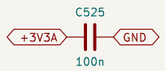




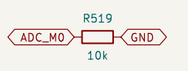


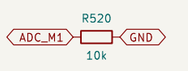


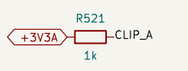














ADC
- U502 makes a differential signal: 1 Vrms at PRE_OUT = 2 Vrms differential = 0 dBFS on the CS5361
  (full scale = 1.13 x VA = 5.65 V p−p differential at VA = 5 V).
- U503 CS5361: slave mode, I2S, HPF on. M0/M1 speed mode set by XMOS per sample rate.
  Unused right channel tied to VQ. OVFL drives the clip LED.
- VERIFY: CS5361 FILT+/VQ capacitor values and input RC against the datasheet.

All−in−one audio interface

VincentM9

Sheet: /ADC driver and ADC (CS5361)/

File: adc.kicad_sch

Title: ADC driver and ADC (CS5361)

Size: A3	Date: 2026−10−23	**Rev: A**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 15/15